

Unit 10, Lesson 4: Solving One Step Inequalities Using Multiplication and Division

Benchmark

MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.

Learning Targets

- ☐ I can solve one-step inequalities in one variable using multiplication or division.
- ☐ I can relate properties of inequality to solving one-step inequalities algebraically.

10.4.1 Warm-Up: Bell Work

1. Using substitution, determine whether the given numerical value is part of the solution set of the inequality. Show your work.
 - a. Is $n = -5.5$ part of the solution set for $-4n > 20$?
 - b. Is $y = 8$ part of the solution set for $-6 \leq -\frac{3}{4}y$?
 - c. Is $b = 2.7$ part of the solution set for $-9.6 > -12.3 + b$?
2. Explain how to determine whether a value is part of the solution set for an inequality.

10.4.2 Collaboration: The Effect of a Negative Coefficient

1. Two simple inequalities are shown.

$$1 > 0 \quad \text{and} \quad -1 > 0$$

- a. Circle the inequality that is true and cross out the inequality that is false.
- b. Complete the statement.

If 1 is greater than 0, then the opposite of 1 is

_____ than 0.

- c. Write the correct inequalities by filling in a comparison symbol on each line.

$$0 \text{ ______ } 1 \quad \text{and} \quad 0 \text{ ______ } -1$$

- d. Verify your answers with your partner on parts A-C and, if necessary, resolve any differences.

2. Work with your partner to complete the following.

- a. Complete the inequalities in the table based on your understanding of opposites.

Given Inequality	Equivalent Inequality
$0 < -q$	$0 \text{ ______ } q$
$0 > -m$	$0 \text{ ______ } m$
$-w \leq 0$	$w \text{ ______ } 0$

In each equivalent inequality, the sign of the variable changed to its opposite and the value 0 remained the same.

- b. Describe which operation(s) could be applied to both sides of the given inequalities above to result in the signs of the variables changing and the value 0 remaining the same.

- c. Describe what happens to the comparison symbol in the inequality when either of these operations are applied.

10.4.3 Bring It Together: The Effect of a Negative Coefficient

1. Complete the statements using words from the bank shown. Some words may not be used.

adding

subtracting

multiplying

dividing

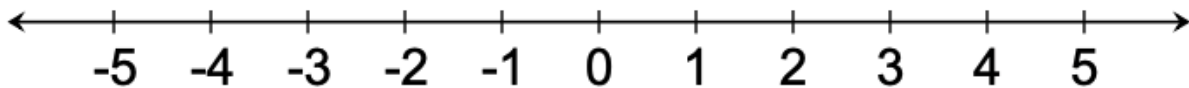
opposite

When the variable term in an inequality is negative, the inequality can be rewritten by _____ or _____ the inequality by -1 . In the resulting inequality, each term and the comparison symbol will change to its _____.

2. Use the inequality $-n \geq 2$ to complete the following.
- a. Complete the inequality to create an equivalent inequality.

 n _____

- b. Graph the solution set for $-n \geq 2$.



10.4.4 Guided Instruction: Solving One-Step Inequalities with Multiplication and Division

As with solving one-step inequalities with addition and subtraction, properties of inequality can be used to solve inequalities with multiplication and division. As discovered in the Collaboration, though, when the value that is being multiplied or divided is negative, this changes both the values in the inequality *and* the comparison symbol to their opposites.

Multiplying both sides of an inequality by the same value can be used to solve an inequality by the ***multiplication property of inequality***.



Multiplication Property of Inequality

If $a > b$ and $c > 0$, then $a \times c$ $b \times c$.

If $a > b$ and $c < 0$, then $a \times c$ $b \times c$.

1. Find the solution to both inequalities using the multiplication property of inequality.

$$\frac{y}{4} \leq 2.1$$

$$\frac{y}{-4} \leq 2.1$$

Dividing both sides of an inequality by the same value can be used to solve an inequality by the ***division property of inequality***.

***Division Property of Inequality***

If $a > b$ and $c > 0$, then $a \div c$ ____ $b \div c$.

If $a > b$ and $c < 0$, then $a \div c$ ____ $b \div c$.

2. Find the solution to both inequalities using the division property of inequality.

$$-25 > 5c$$

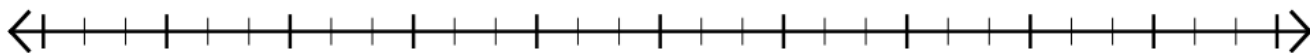
$$-25 > -5c$$

10.4.5 Guided Practice: Solving One-Step Inequalities with Multiplication and Division

1. Use the inequality $7 \geq -\frac{3}{2}y$ to complete the following.

a. Solve the inequality. Show your work.

b. Graph the solution set on the number line.

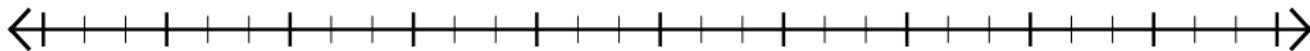


c. Use substitution to test a value or values to verify that the solution and the graph are correct.

2. Use the inequality $\frac{k}{3} < -5$ to complete the following.

a. Solve the inequality. Show your work.

b. Graph the solution set on the number line.



c. Use substitution to test a value or values to verify that the solution and the graph are correct.

3. Gabriel and Lauren made errors when solving the inequality $-4x \leq 12$. Their work is shown.

Gabriel's Work	Lauren's Work
$\begin{array}{r} -4x \leq 12 \\ +4 \quad +4 \\ \hline x \leq 16 \end{array}$	$\begin{array}{r} -4x \leq 12 \\ \hline -4 \quad -4 \\ \hline x \leq -3 \end{array}$

- a. Explain which mistake(s) each student made.

Gabriel's Error	Lauren's Error

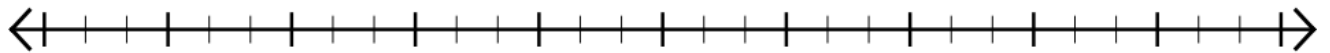
- b. Find the correct solution to $-4x \leq 12$.

10.4.6 Your Turn!

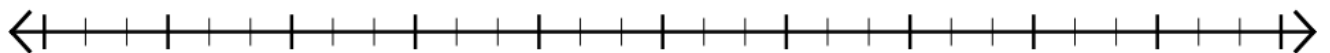
1. Frankie got a summer job as a lifeguard at the local pool, where he will earn \$9.50 per hour. He wants to make at least \$299 to buy the newest gaming system. The inequality $299 \leq 9.50x$ mathematically represents this situation where x is the number of hours Frankie works. Find the solution to the inequality to determine the minimum number of hours Frankie must work to save enough money for the gaming system. Round your answer to the nearest hundredth.

2. Solve each inequality. Then, graph its solution set.

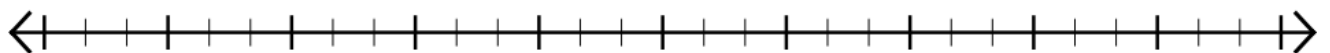
a. $9x > -\frac{3}{4}$



b. $5.5 \leq -\frac{n}{8}$



c. $-3.6y \leq -17.1$



10.4.7 Wrap-Up: Cool Down

Two inequalities are shown.

$2.5x \geq -10$

$-2.5x \geq -10$

1. Complete the table to describe how the process of solving each inequality is similar and different.

How is the process of solving each inequality similar?	How is the process of solving each inequality different?

